

|                                                        |                                                     |                                                 |                                       |                                                |
|--------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------|---------------------------------------|------------------------------------------------|
| This is sample data provided for CALIBER purposes only |                                                     | <b>ONE POINT LESSON (OPL)</b>                   |                                       | OPL No: <b>OPL-DC-3401A-06</b><br>Sheet 1 of 1 |
| <b>OPL Title</b>                                       | <b>Multipoint Thermocouple TE-3401 Verification</b> | <b>Discipline</b>                               | Instrument                            |                                                |
| <b>Equipment</b>                                       | DC-3401A - CATALYST REDUCTION REACTOR               | <b>Area / Unit</b>                              | 3400 - CATALYST REDUCTION SYSTEM      |                                                |
| <b>Related Interlock</b>                               | SEQ-3401 (SIL 2)                                    | <b>P&amp;ID Ref</b>                             | TJC-LLD-PID-3401                      |                                                |
| <b>Classification:</b>                                 | <input type="checkbox"/> Basic Knowledge            | <input checked="" type="checkbox"/> Improvement | <input type="checkbox"/> Trouble Case | Aspect: P-Q-C-D-S-M-E                          |

## 1. PURPOSE / OBJECTIVE

This One Point Lesson explains **multipoint thermocouple te-3401 verification** for DC-3401A (CATALYST REDUCTION REACTOR) in the CATALYST REDUCTION SYSTEM. The 8-point bed temperature profile drives the whole reduction; a failed or drifting thermocouple can hide a hot spot. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

## 2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **N2 / H2 / Pd catalyst** at up to 6 barg / FV / 250 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

## 3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

## 4. DETAILED PROCEDURE / STEPS

| Step | Action                                                             | Check / Acceptance             |
|------|--------------------------------------------------------------------|--------------------------------|
| 1    | Verify all 8 TE-3401 points are reading and note the spread        | Loop isolated / under permit   |
| 2    | Flag any single TC more than 15 degC off its neighbours            | Reference standard traceable   |
| 3    | Cross-check the profile against TIC-17343 heater outlet            | As-found & as-left recorded    |
| 4    | Replace any open-circuit or drifting TC at the next opportunity    | Trip / control action verified |
| 5    | Confirm TSHH-3401 selects the highest bed temperature for the trip | Return to service confirmed    |

## 5. COMMON PROBLEMS & TROUBLESHOOTING

| Symptom                                             | Likely Cause                                           | Action                                                            |
|-----------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------|
| Bed multipoint TC point 3 reading open circuit      | Thermocouple element failed at the sheath junction     | Replaced TC point 3 in the multipoint assembly, verified reading. |
| Inerting O2 analyzer drift vs portable reference    | Analyzer zero drift and sample dryer saturated         | Recalibrated AI-3401 on span gas, replaced sample dryer           |
| N2 inert valve slow / sticking during inerting step | Actuator seals hardened, positioner out of calibration | Serviced actuator, replaced seals, recalibrated positioner        |

## 6. KEY LEARNING POINTS

- The 8-point bed temperature profile drives the whole reduction; a failed or drifting thermocouple can hide a hot spot.
- Respect the interlock SEQ-3401: e.g. TSHH-3401 trips at > 230 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-3401 before working on DC-3401A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

| Prepared by                 | Reviewed by (Supervisor)   | Approved by (Manager)    | Date of Sharing     |
|-----------------------------|----------------------------|--------------------------|---------------------|
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